

Figure 1: Accessibility requirements

the adaptability and affordability of Free and Open Source Software (FOSS) can make a life-changing impact on people with special needs.

Accessibility requirements

Disabilities can be classified as visual, auditory, etc. The accessibility tools for each type of disability are of a specialised type. The specific requirements for each type of disability are illustrated in Figure 1.

A complete list of all the available accessibility tools is beyond the scope of this article. Hence the tools mentioned in this article are representative in nature, for each category, and the list is by no means exhaustive. The tools have been chosen based on my own personal experiences and discussions with users with special needs.

For the visually challenged

In the case of visual impairments, text-to-speech conversion tools play a major role. As the visually challenged cannot view the contents on the screen either partially or completely, they require tools to convert text into speech. Audible alerts are audio indications for some of the actions either performed by the system or the user, like connecting to a network, receiving a mail, etc. Most current operating systems include audible alert features, which are located in *Accessibility options*.

Screen readers

Screen readers are tools that present an audio version of the screen's contents to the visually challenged. Here is a list of free and open source screen readers (not in any specific order):

- **NVDA** stands for *Non Visual Desktop Access*. This free and open source screen reader is becoming very popular in the visually challenged community. Configuring this screen reader is not difficult. The

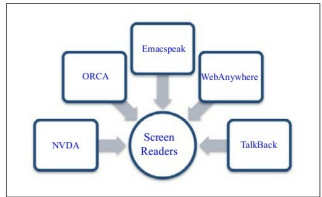


Figure 2: Screen readers

recent version of NVDA 2013.3 can be downloaded from <http://www.nvaccess.org/download/>. NVDA has the power to support Braille displays as well. Another major feature of NVDA is the ability to run it from USB drives, which gives it portability. One of its limitations is that it only supports Windows. Being a FOSS tool, if you wish to contribute to NVDA by enhancing its functionalities or translating it to another language, visit <http://community.nvda-project.org/>.

- **ORCA**: In the Linux environment, one of the most widely used screen readers is ORCA. If you are using Linux distributions like Ubuntu or Fedora, you need not install ORCA as it comes as a part of GNOME 2.16, or later versions. In case you are on an older version or distribution that doesn't have ORCA by default, it can be downloaded from <https://wiki.gnome.org/Projects/Orca/DownloadInstall>. The speech synthesiser behind ORCA is *eSpeak*. Constant efforts are being made to enhance *eSpeak* to produce a more human voice rather than the metallic sounding robotic voice. ORCA provides support for Braille through *BrITTY* and *liblouis*.
- **Emacspeak**: The screen readers simply read the text that appears on the screen. Emacspeak raises the bar one level higher by making an attempt to provide sensible *eyes-free access*. The brain behind the development of Emacspeak is T V Raman, who himself has been visually challenged since childhood. He is presently leading the accessibility development for Google Chrome and Android. Emacspeak doesn't simply read content but provides it in a more meaningful manner. With the growing number of features available on Web pages, Emacspeak is a powerful tool for the visually challenged to harness the full power of the Internet. Emacspeak can be downloaded from <http://emacspeak.sourceforge.net/>.
- **WebAnywhere**: The screen readers listed above need some sort of installation. If you want a screen reader that doesn't require any installation at all, then WebAnywhere would be the right choice. It can be activated by simply surfing to <http://webanywhere>.